STEM Stop

Day 2 - Biology

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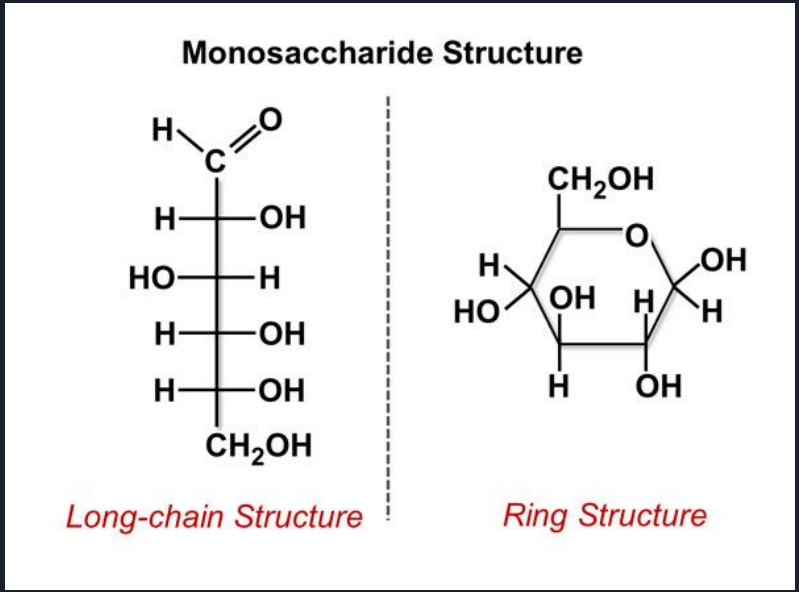
Macromolecules

Carbohydrates

- Function: Provides your body with quick energy
- Monomer: Monosaccharide
- Structure:

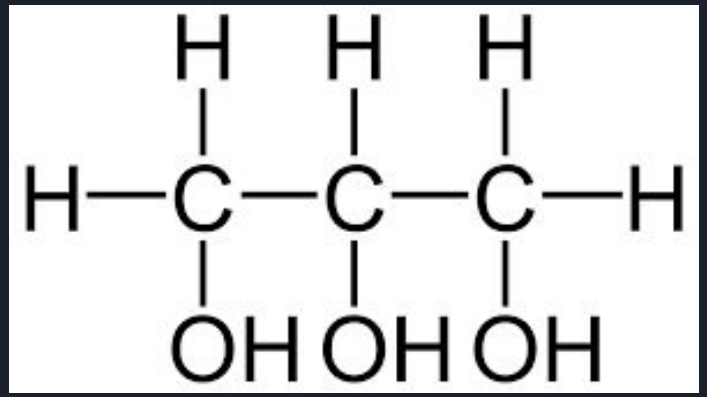


Carbohydrates

- Only Contain the elements Carbon, Hydrogen, and oxygen in a ratio of 1:2:1 ($C_6H_{12}O_6$)
- Monosaccharides (one sugar) can be identified by their sweet taste
  - Glucose
  - Fructose
- Disaccharides (2 sugars) are a combination of 2 monosaccharides
  - Glucose + Glucose = Maltose
  - Glucose + fructose = Sucrose
- Polysaccharides are made of long complex carbohydrate chains (3 or more monosaccharides)
  - Starches (Bread,Cereal,Pasta)
  - Cellulose(plant cell walls)
  - Chitin (Fungi cell walls and insect exoskeleton)

Lipids

- Function: Long term energy,hormones,steroids,cell membrane, insulation, and protection
- Monomer: Triglyceride (1 glycerol and 3 fatty acids)
- Structure:

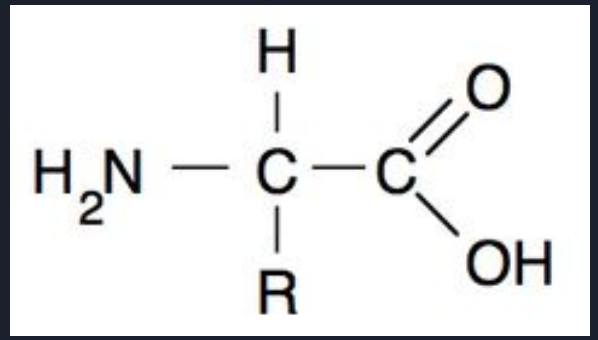


Lipids

- It is made up of Carbon, Hydrogen, and Oxygen and can’t dissolve in water
- Either saturated or unsaturated
- Saturated
  - Solid at room temperature
  - Considered “Bad fats”
  - Butter, lard, grease,etc.
- Unsaturated
  - Liquid at room temperature
  - Better for you
  - Olive Oil, Fish oil, vegetable oil,etc.

Proteins

- Function: Helps with the organ's structure, function and regulation. Creates enzymes to control rate of chemical reaction and builds muscle
- Monomer: Amino Acids
- Structure:



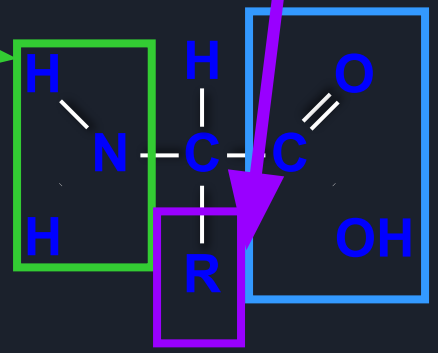
Proteins

- They are made up of Carbon, Oxygen, Hydrogen, Nitrogen, and sometimes Sulfur

and an "R" group which varies in the different amino acids. This is why there are 20 different amino acids.

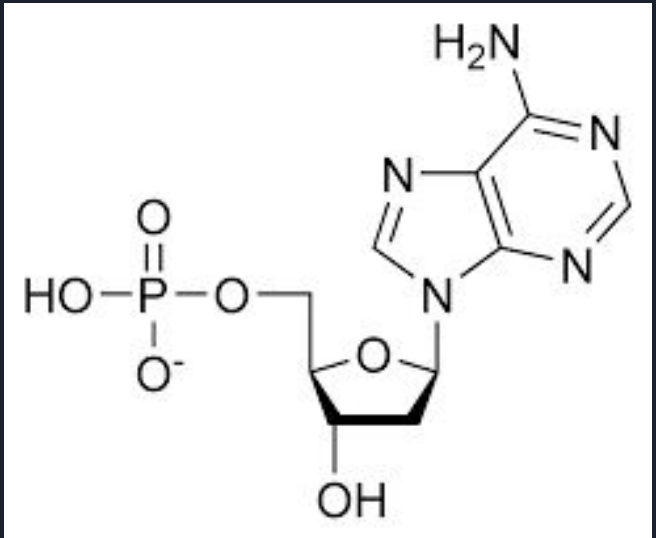
an amino group

a carboxyl group



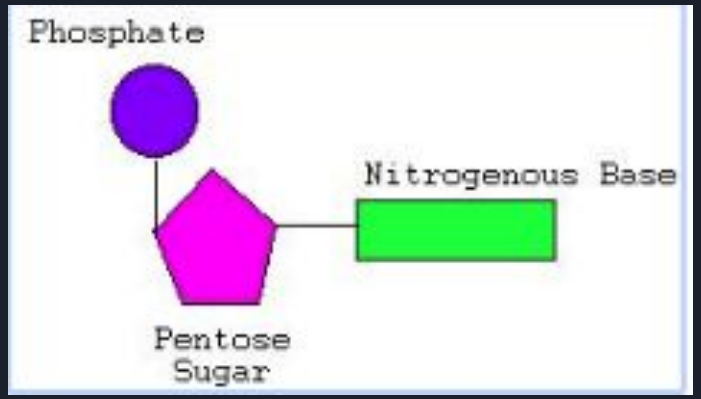
Nucleic Acids

- Function: Store genetic information, help make proteins
- Monomer: Nucleotide
- Structure:



Nucleic Acids

- Contains the elements carbon, hydrogen, nitrogen, oxyten, and phosphorous
- Contains a sugar,phosphate, and a N-base
- Only macromolecule that can replicate itself

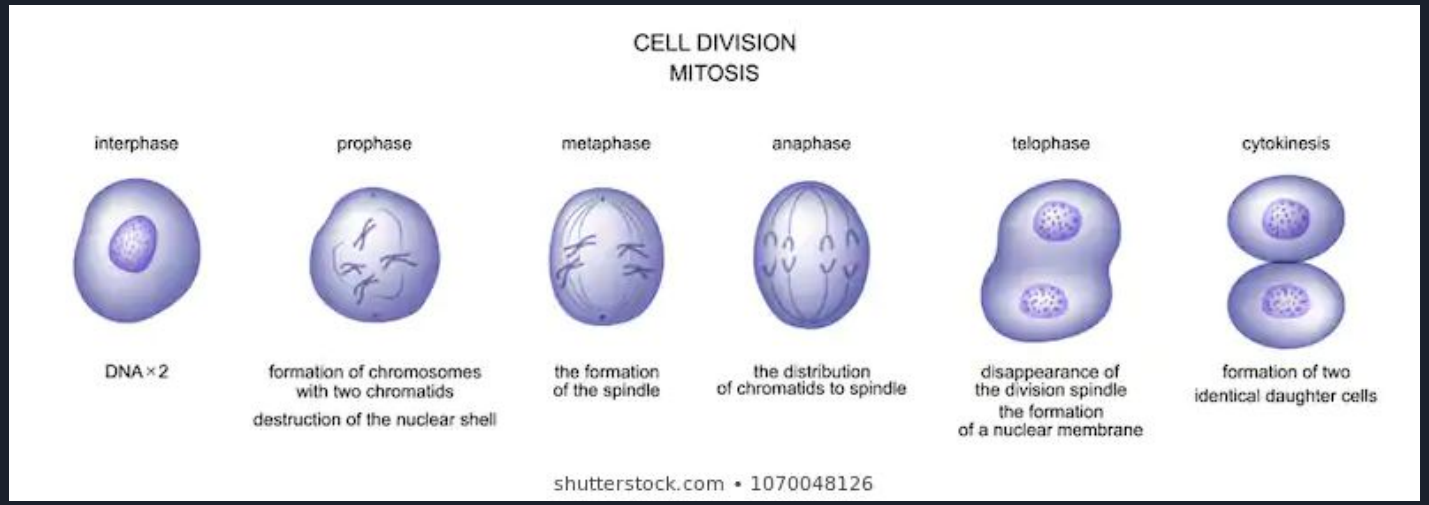


Questions 1

Genetics

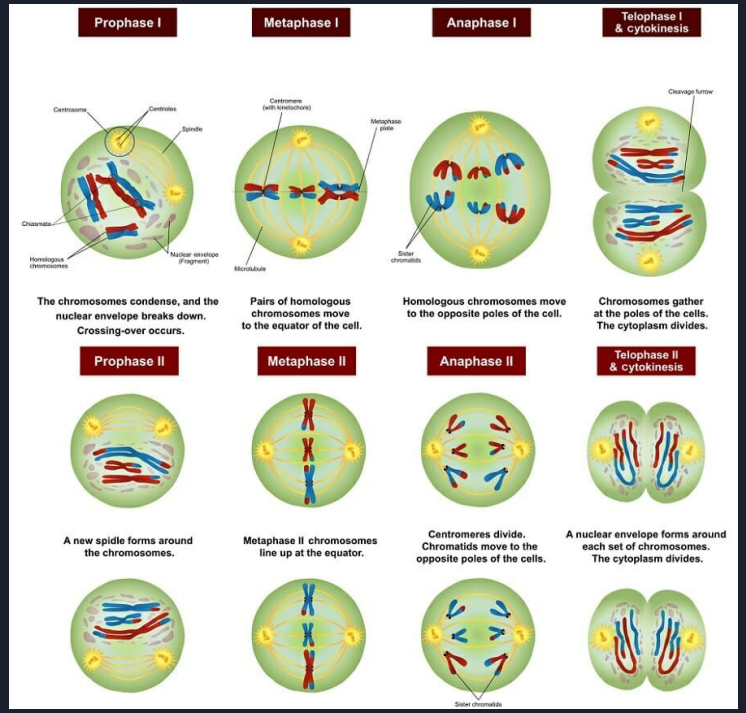
Mitosis

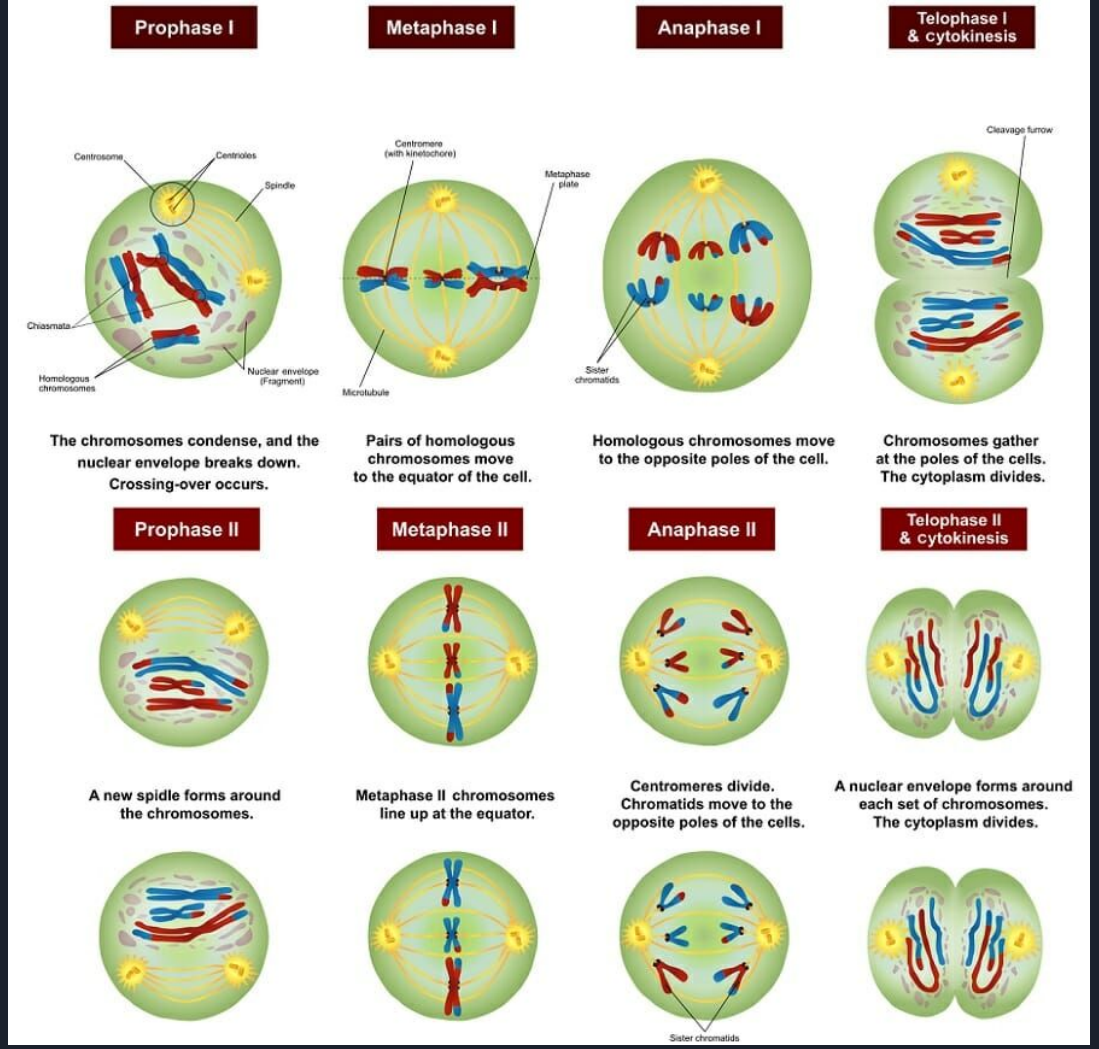
- Mitosis is the process of a cell duplicating its nucleus and splitting into 2 identical cells
- The main steps are:



Meiosis

- Meiosis is the process of creating sex cells a.k.a gamete cells




Prophase I
Metaphase I
Anaphase I
Telophase I & cytokinesis
Centrosome
Centrioles
Spindle
Chiasmata
Homologous chromosomes
Nuclear envelope (Fragment)
Centromere (with kinetochore)
Metaphase plate
Microtubule
Sister chromatids
Cleavage furrow
The chromosomes condense, and the nuclear envelope breaks down. Crossing-over occurs.
Pairs of homologous chromosomes move to the equator of the cell.
Homologous chromosomes move to the opposite poles of the cell.
Chromosomes gather at the poles of the cells. The cytoplasm divides.
Prophase II
Metaphase II
Anaphase II
Telophase II & cytokinesis
A new spidle forms around the chromosomes.
Metaphase II chromosomes line up at the equator.
Centromeres divide. Chromatids move to the opposite poles of the cells.
A nuclear envelope forms around each set of chromosomes. The cytoplasm divides.
Sister chromatids

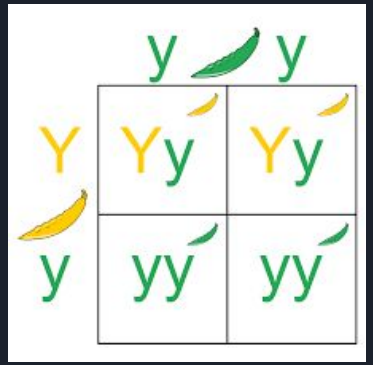
Questions 2-3

Gregor Mendel

- Gregor Mendel is known as the “Father of Genetics”
- He was a monk that ran several experiments involving pea plants and seeing how the features of the parents were passed onto their offspring

Punnett Squares

- Punnett Squares are a form of predicting the likelihood of an offspring having a certain trait depending on the parents
- Dominant Traits: These traits will be expressed whenever the organism has it and is denoted with capital letters on the Punnett Square
- Recessive Traits: These traits will be only be expressed when an organism doesn't have a conflicting dominant trait and is denoted with lowercase letters



Multiple Alleles, Incomplete, and Codominance

- Multiple Alleles: This is when there are more than 2 genotypes for an organism
  - Human Blood is A, B, O
- Incomplete Dominance: This is when the dominant and recessive traits of a heterozygous trait gets blended together
  - Red Flower + White Flower = Pink Flower
- Codominance: This is when the dominant and recessive traits of a heterozygous trait both are expressed separately
  - White Flower + Black Flower = Spotted Flower

Double Punnett Squares
Dihybrid cross

- Double Punnett Squares are used to predict the likelihood of 2 traits instead of one
- Here is an example of RrYy X RrYy:

	RY	***Ry***	***rY***	***ry***
RY	*RRYY*	*RRYy*	*RrYY*	*RrYy*
Ry	*RRYy*	*RRyy*	*RrYy*	*Rryy*
rY	*RrYY*	*RrYy*	*rrYY*	*rrYy*
ry	*RrYy*	*Rryy*	*rrYy*	*rryy*

Questions 4-10

Break

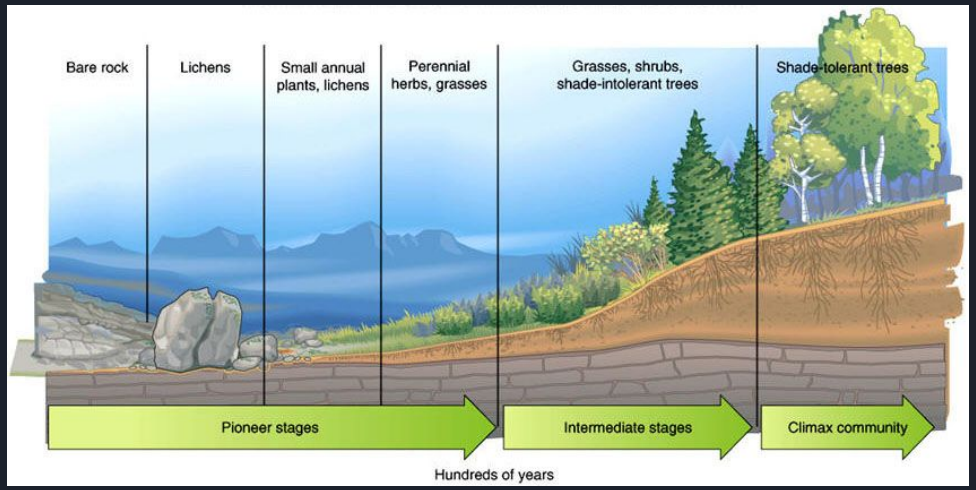
Succession

What is Succession?

- Succession is the process of life growing back into an environment that was previously uninhabited or destroyed

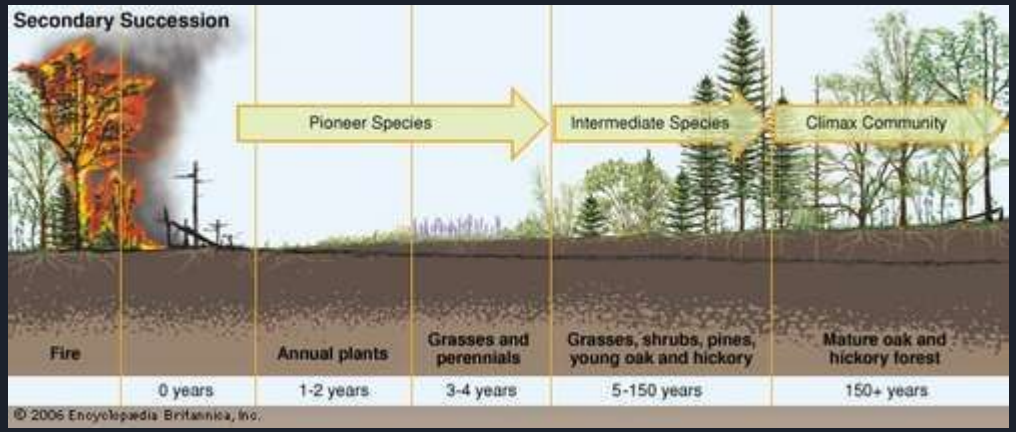
Primary Succession

- Primary Succession is when life grows onto a place that never had living things beforehand
- Examples are: Newly exposed rock in a glacier, new rock formed by volcanoes,etc.



Secondary Succession

- Secondary Succession occurs when something reduces an already established ecosystem and life regrows and recovers
- Life can be wiped out due to fires, floods, hurricanes, etc.
- Biggest difference from primary is that there is still soil in the ecosystem

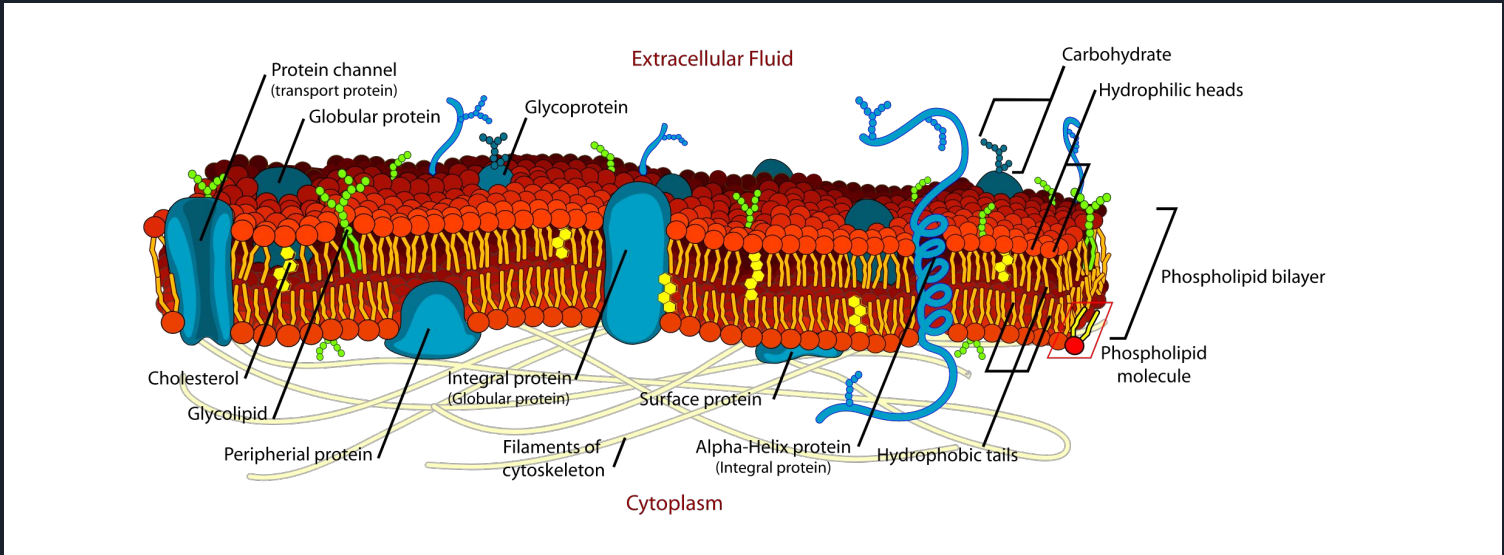


Questions 11-12

Cell Transport

Cell Membrane

Function : regulates what enters and leaves the cell



Structure: a double layered sheet called a lipid bilayer

Function

Lipids: there are two layers of lipids → bilayer

Proteins: form gates, channels or pumps that move material into or out of the cell

Carbohydrate chains: cell identification, so that cells can recognize each other

Transport

- **Permeable**: membrane allows substances to pass through
  - ex. Water through a strainer

- **Selectively permeable or semi-permeable:** only allows certain molecules to pass through
  - molecules cannot pass through if they are too big, or strongly charged
  - Spaghetti cannot pass through strainer, it is too big
  - screen on a window: fresh air can pass through, but most bugs cannot

Equilibrium

- when particles pass through the membrane without any added energy
- particles are in constant motion back and forth across the membrane
- there is no net change in concentration

Concentration Gradient

STEM Stop

Go with the flow and move from **HIGH** to **LOW**

Two Ways molecules pass through membranes

1) <u>PASSIVE TRANSPORT</u> - molecules pass through the cell membrane without the use of energy

3 Types of Passive Transport across cell membrane

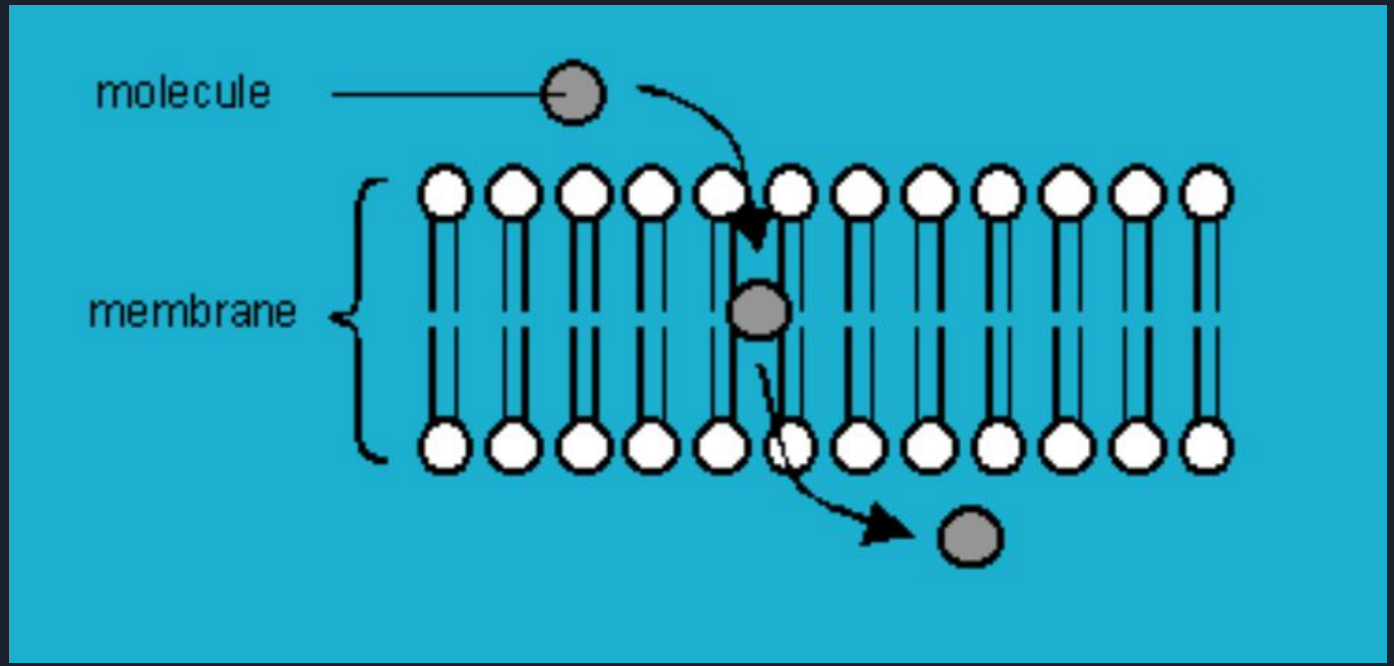
1. Diffusion
2. Facilitated Diffusion
3. Osmosis

Diffusion

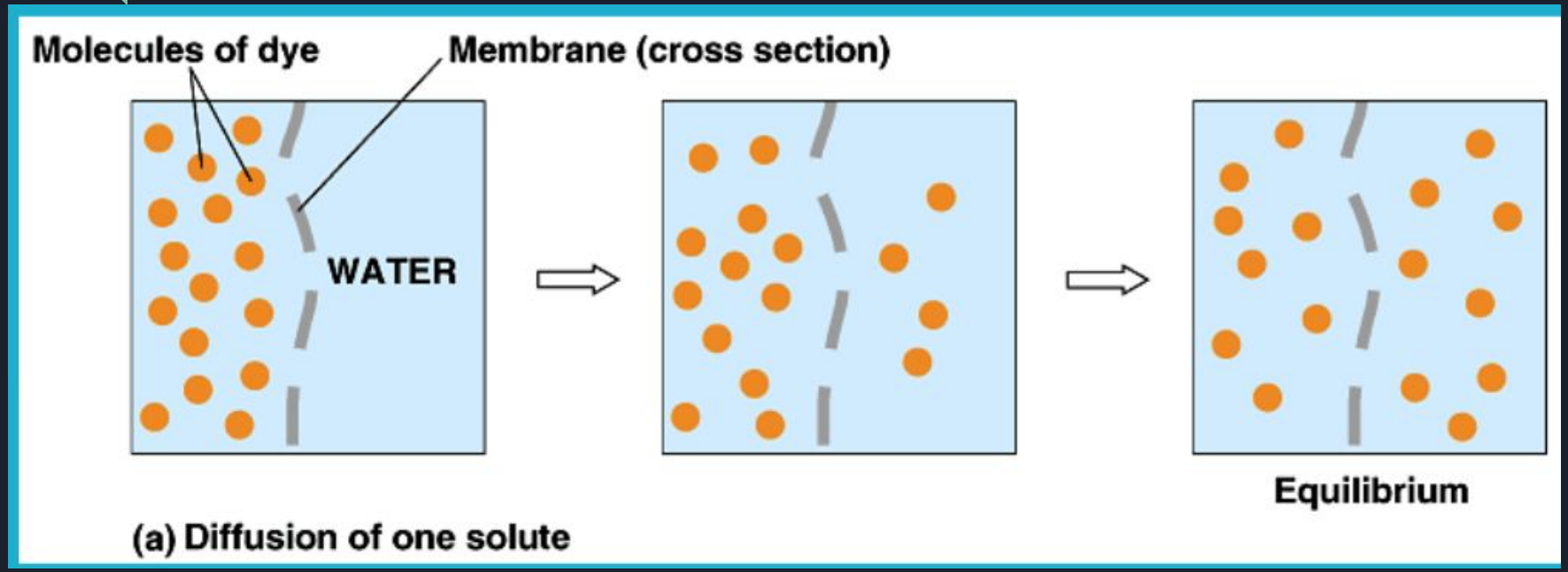
process where molecules of a substance move from high concentration to low concentration (down concentration gradient)

- <u>Examples</u>: Food coloring in a beaker of water, perfume in a corner of the room

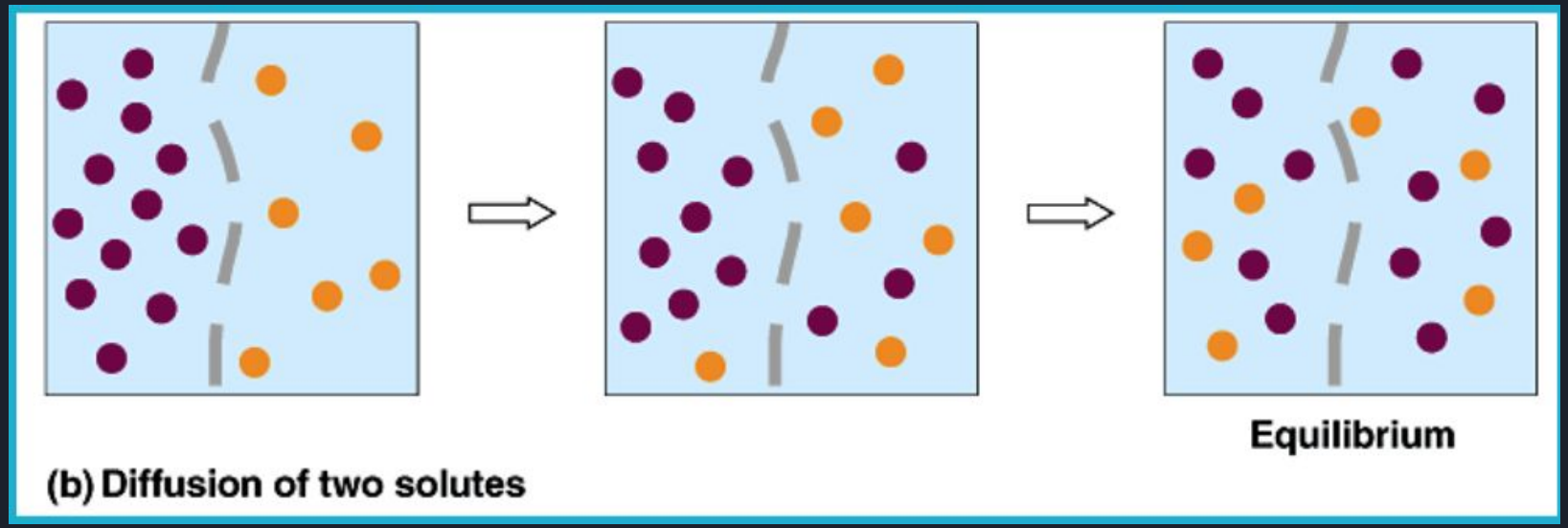
Passive Transport



Diffusion


Molecules of dye
Membrane (cross section)
WATER
Equilibrium
(a) Diffusion of one solute

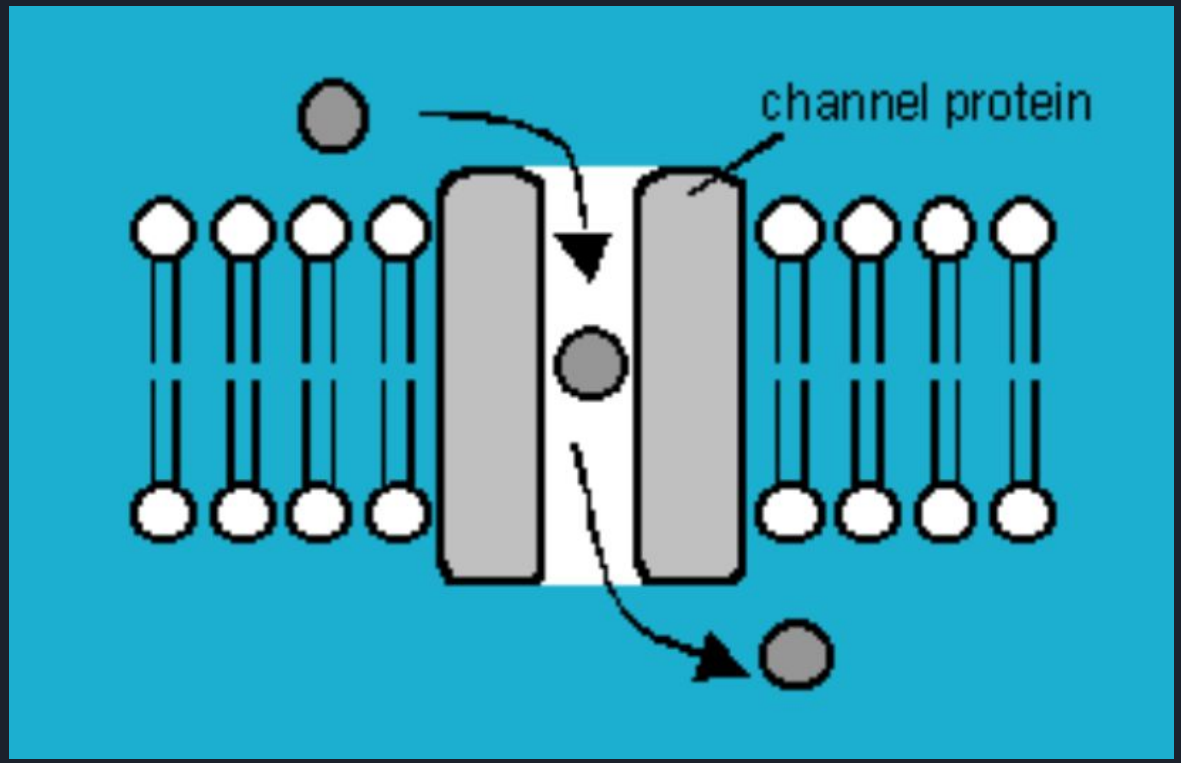
Diffusion



Facilitated Diffusion

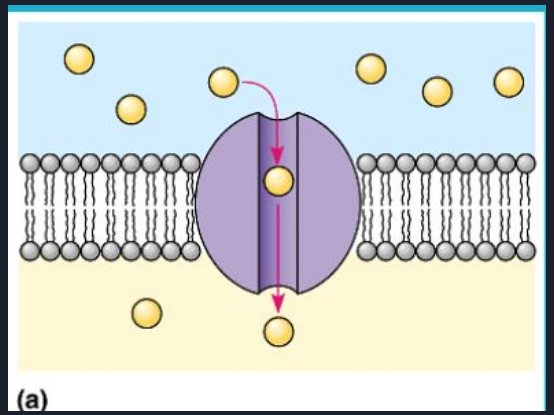
when a cell membrane protein facilitates or helps a molecule diffuse through the membrane

Facilitated Diffusion



Types of Facilitated Diffusion

- Ion channel – transport protein with a polar pore through which ions can pass
- Thus, an ION can cross the cell membrane without contacting the nonpolar interior of the lipid bilayer
  - Ions: K^+, Cl^-, Na^+, Ca^+

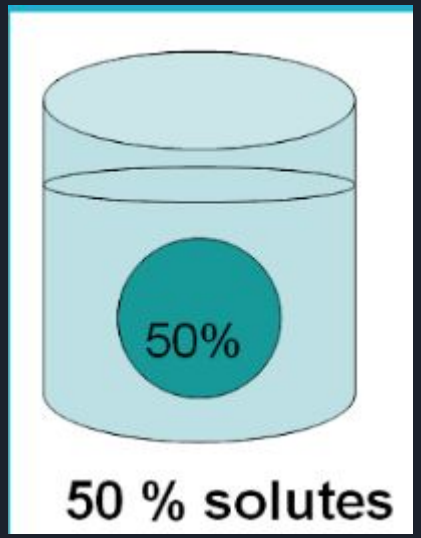


Osmosis

- The **diffusion of water** through a selectively permeable membrane
- Cell membranes are completely permeable to water, therefore, the environment the cell is exposed to can have a dramatic effect on the cell.

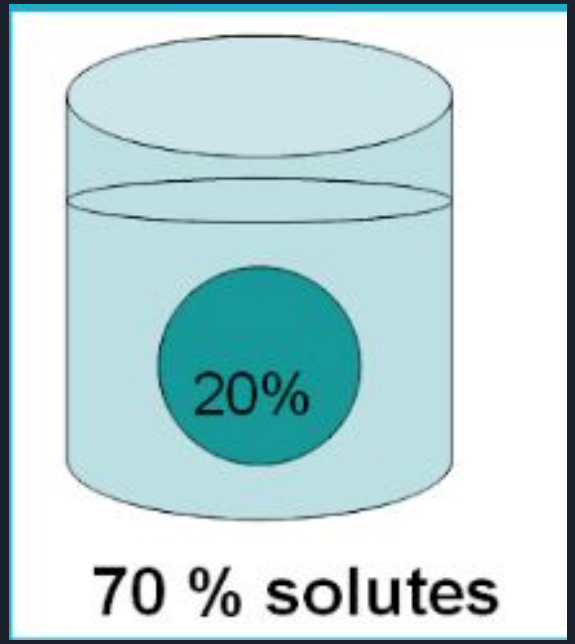
Isotonic

means “same strength” when the concentration of each side of the membrane is the same



Hypertonic

“above strength” solution has a higher solute solution than the cell

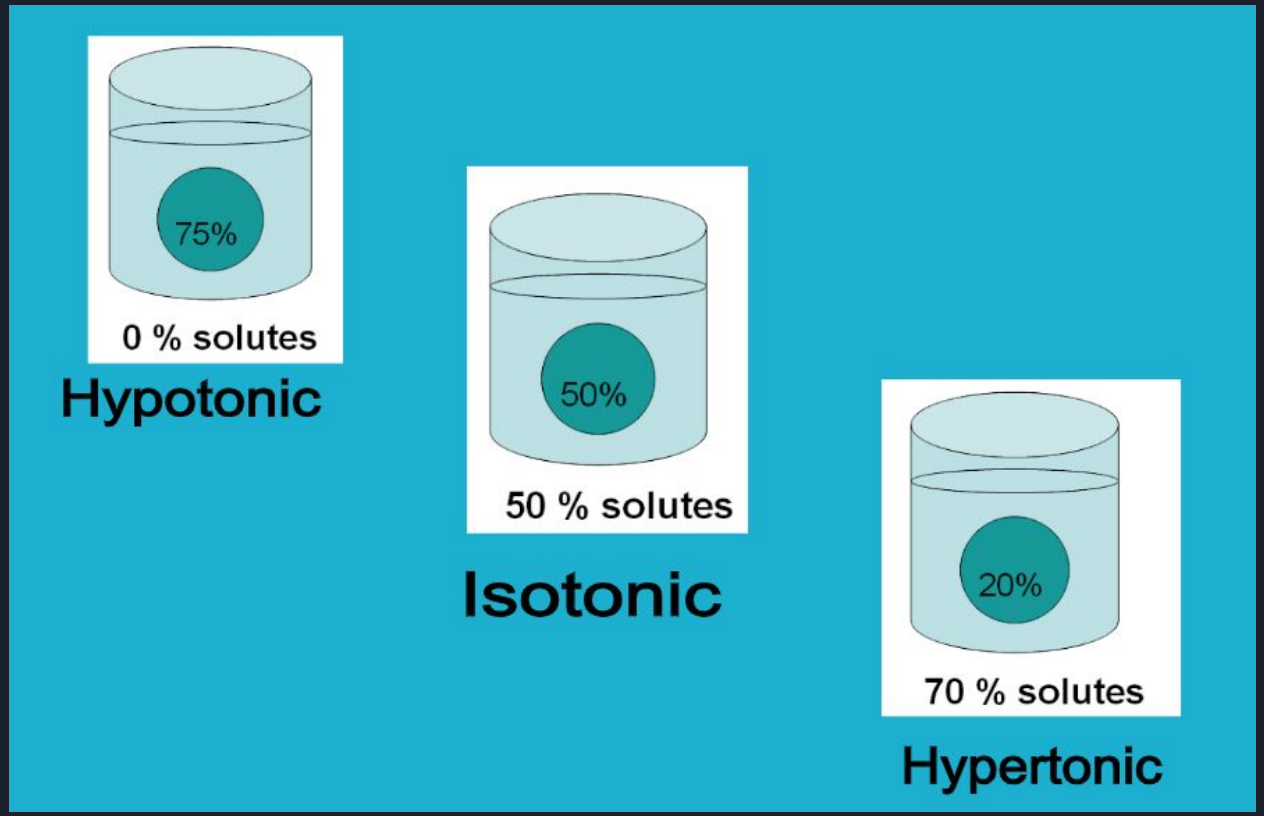


STEM Stop

Hypotonic

“below strength” solution has a lower solute solution than the cell

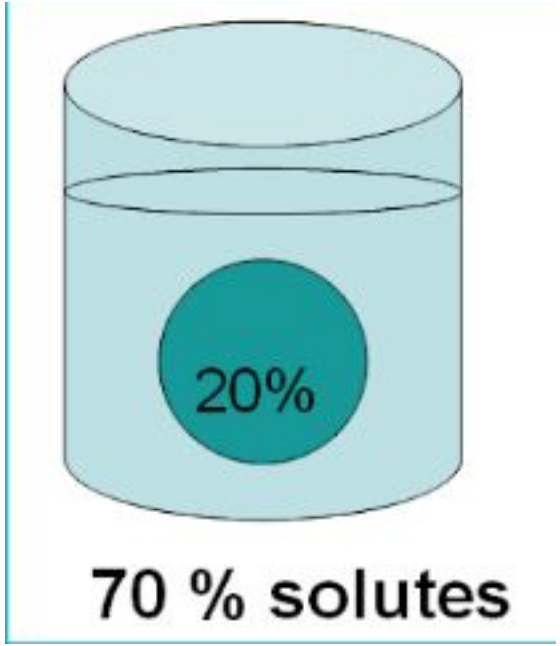
Examples

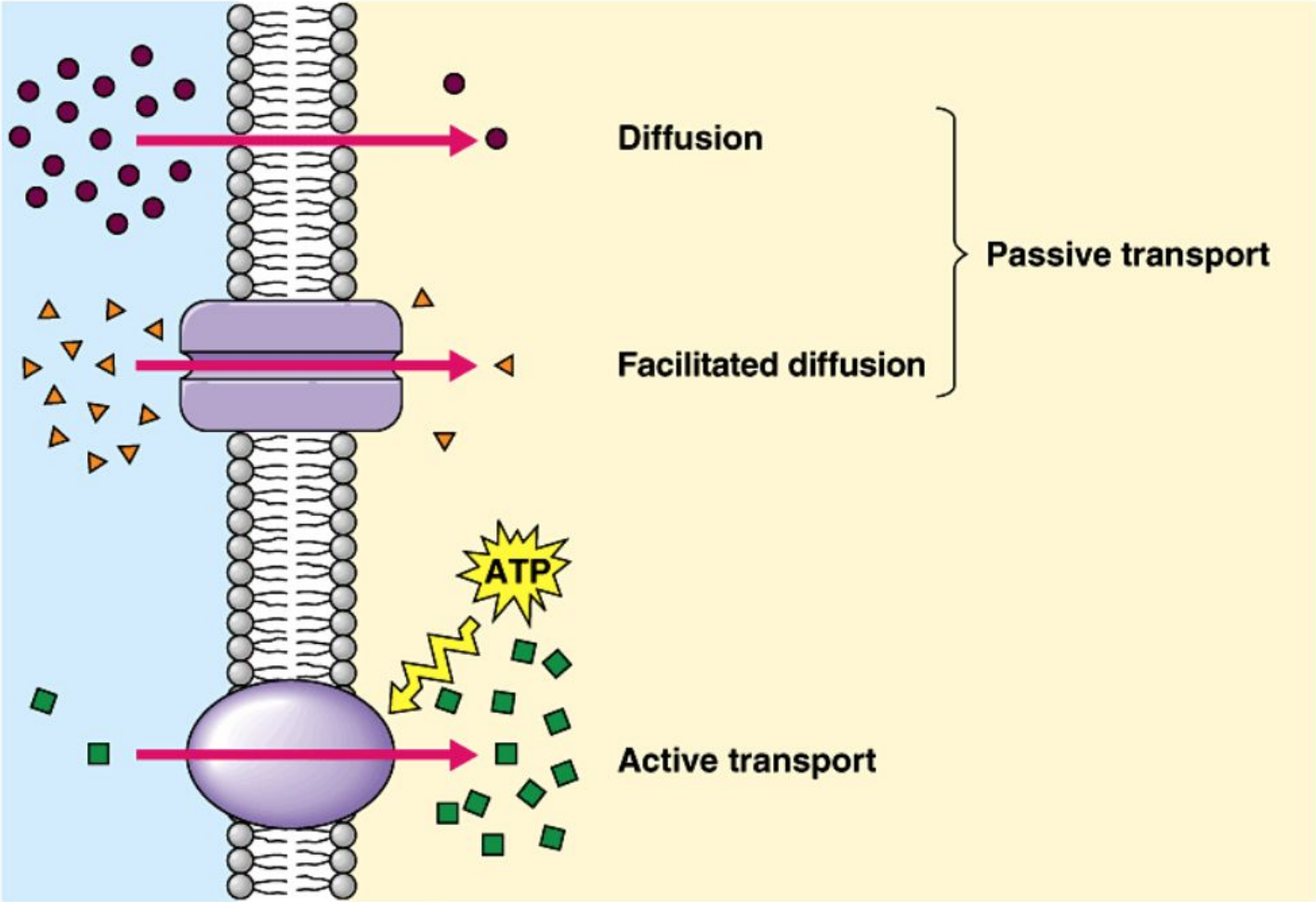

75%
0 % solutes
Hypotonic
50%
50 % solutes
Isotonic
20%
70 % solutes
Hypertonic

Questions 13-25

Active Transport

energy is necessary for molecules to pass through the cell membrane





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Active Transport

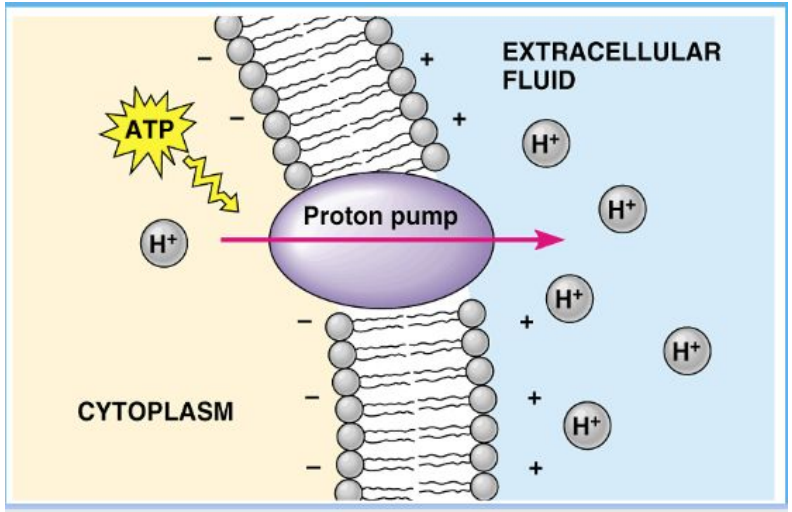
- uses the cell's energy to move molecules <u>against</u> a concentration gradient
- proteins pump ions and molecules across the membrane from low to high concentration

Protein Pump

- A membrane protein pumps molecules across the membrane against the concentration gradient (low→ high)
  - **<u>USES ENERGY</u>**

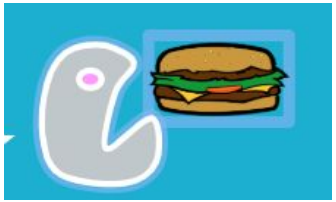
Active Transport - Ions

- <u>Proton</u> <u>Pump</u>
  - H+ (protons) are pumped across the membrane from low to high concentration with help of ATP



Endocytosis

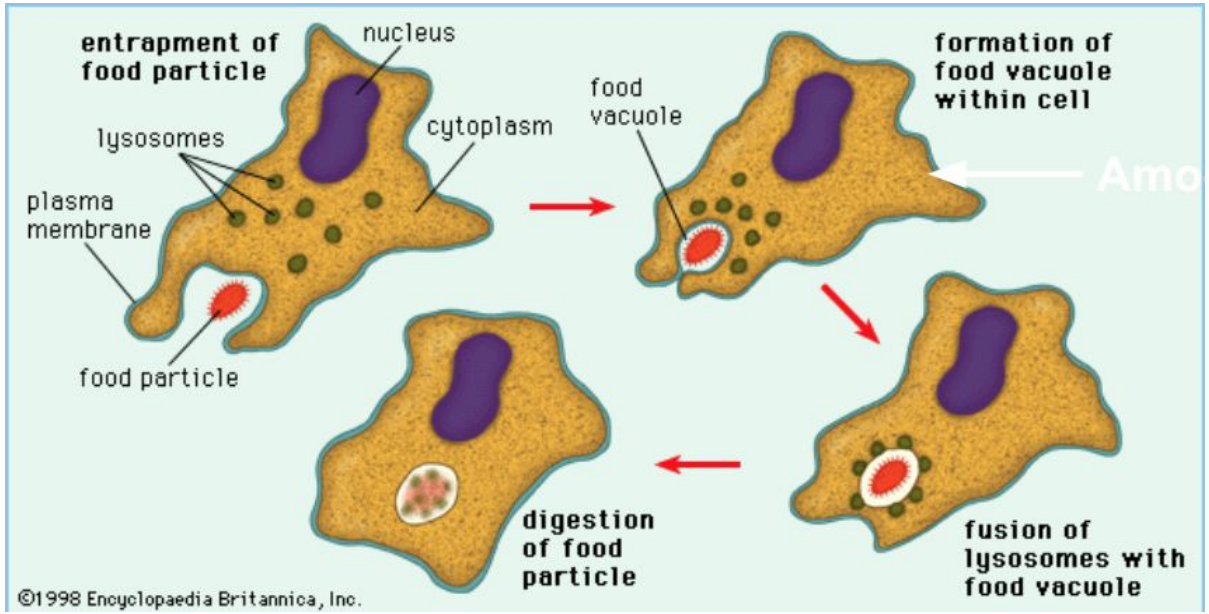
- Bringing large particles into the cell using the cell membrane
  - There are 2 types:
    - Phagocytosis



    - Pinocytosis ---->

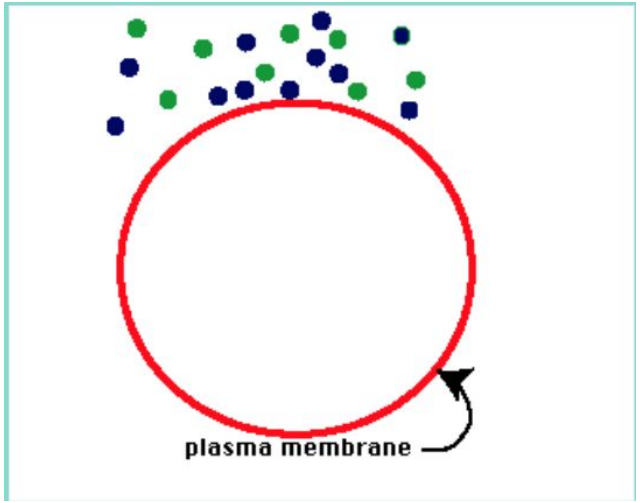
Phagocytosis

- This is the ingestion of large particles.



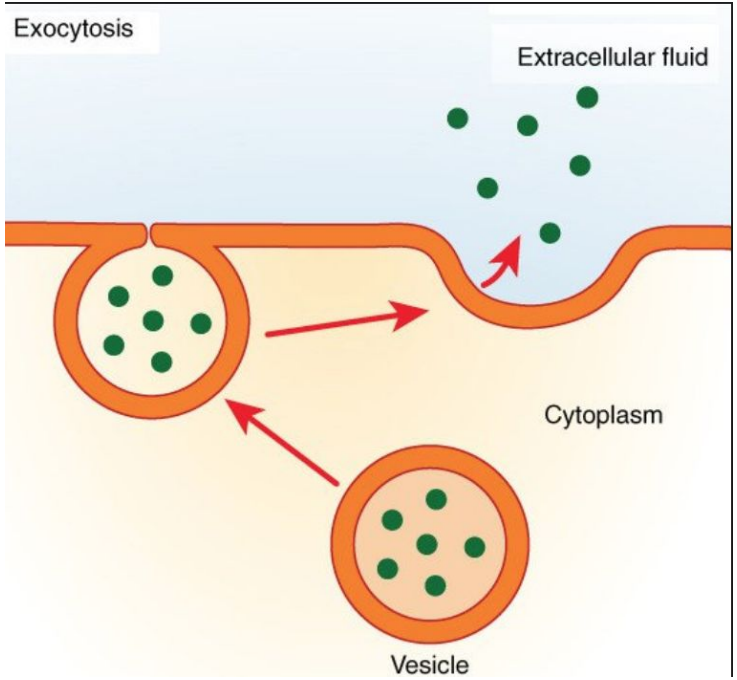
Pinocytosis

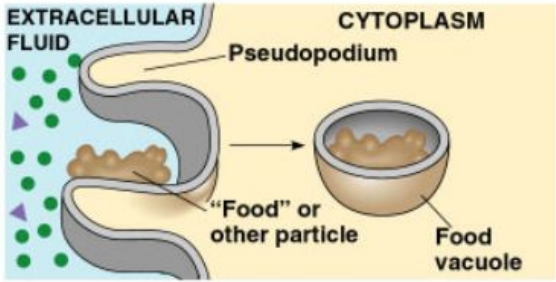
Tiny pockets form along cell membrane, and pinch off into vacuoles inside the cell. Sometimes called “Cell Drinking”



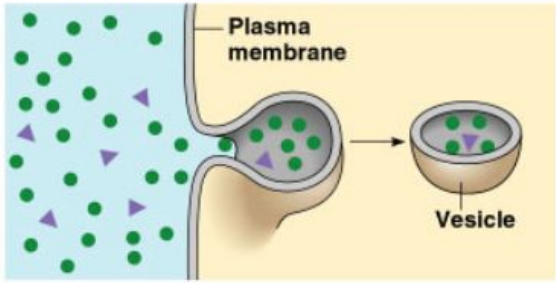
Exocytosis

- This is how the cell gets rid of waste.

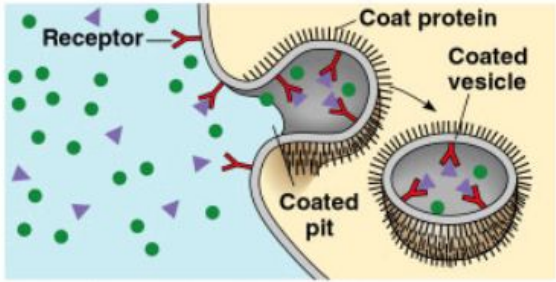




(a) Phagocytosis



(b) Pinocytosis



(c) Receptor-mediated endocytosis

Questions 26-29